

Space-Guard

Autonomous Real-Time Orbital Conjunction Screening & Collision Avoidance

Smart India Hackathon 2026 — Software Edition · Problem ID: 17

Team Space-Guard

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Abstract

This document provides the complete slide-by-slide presentation script, slide visual structures, mathematical and computer science implementation breakdown, team roles, target audience analysis, and tentative development roadmap for the **Space-Guard** platform at the Smart India Hackathon (SIH) 2026. Space-Guard is a deterministic, physics-first orbital conjunction assessment and collision avoidance engine designed for operational spacecraft and mega-constellations.

Contents

1	Slide 1: Title & Team Overview	2
2	Slide 2: Problem Statement — The Low Earth Orbit Crisis	3
3	Slide 3: Solution Statement — How Space-Guard is Better	4
4	Slide 4: Implementation — Physics, Math & Computer Science	5
4.1	1. Physics and Mathematics	5
4.2	2. Computer Science Architecture	6
4.3	3. Satellite Data Acquisition Pipeline	6
5	Slide 5: Live Prototype Walkthrough Plan	7
6	Slide 6: Target Audience & Societal Impact	8
7	Slide 7: Development Roadmap	9

1 Slide 1: Title & Team Overview

Slide 1 — Header & Team Information

Project Name: Space-Guard: Autonomous Orbital Conjunction Assessment & Avoidance

Theme: Space Technology / Smart India Hackathon 2026

Tagline: *Deterministic Physics-First Orbital Defense for Spacecraft & Constellations*

Team Member Contributions

• Team Lead & Full-Stack Software Architect:

- Engineered the asynchronous FastAPI REST microservices architecture and modular routing layer.
- Designed the React 19 single-page application and unified modern component system.
- Integrated Celestrak active TLE ingestion caching pipelines with sub-second response times.

• Orbital Mechanics & Astrodynamics Lead:

- Derived and implemented the SGP4 analytical propagation routines in Geocentric Celestial Reference System (GCRS/ECI).
- Formulated the two-stage screening architecture: $O(N^2)$ apogee/perigee altitude filter and Golden-Section scalar minimization for exact Time of Closest Approach (TCA).
- Integrated the Foster/Alfano 2D Gaussian collision probability (P_c) analytical formulation.

• Control & Maneuver Planner Lead:

- Implemented the Clohessy-Wiltshire (CW / Hill's) linearized relative state transition matrices (Φ_{rr}, Φ_{rv}).
- Developed the Singular Value Decomposition (SVD) optimizer for maximal along-track separation per unit ΔV .
- Formulated the $14\times$ early secular drift fuel efficiency scaling model.

• Data & 3D Visualization Engineer:

- Developed the Three.js WebGL orbital radar engine with accurate inclination (i), RAAN (Ω), and true Earth scale.
- Designed the interactive 2009 collision replay with dynamic debris fragment kinematics.
- Optimized 60 FPS WebGL rendering with asynchronous worker thread decoupling.

2 Slide 2: Problem Statement — The Low Earth Orbit Crisis

The Modern Orbital Environment

- Over **27,000+** cataloged artificial objects larger than 10 cm orbit Earth at velocities exceeding **7.5 km/s (27,000 km/h)**.
- With commercial mega-constellations (Starlink, OneWeb, Project Kuiper), satellite density in Low Earth Orbit (LEO) has surged by over **400%** in five years.

Real-World Case Study: The 10 February 2009 Collision

Historical Ground Truth Benchmark

On **10 February 2009 at 16:56 UTC**, at an altitude of **789 km above the Taymyr Peninsula, Siberia**:

- **Iridium 33** (an active 560 kg commercial communications satellite) collided with
- **Cosmos 2251** (a derelict 900 kg Russian military satellite)
- **Relative Collision Velocity: 14.12 km/s (50,832 km/h)** — a near head-on intersection.
- **Consequence:** Both spacecraft were obliterated in milliseconds, generating over **2,000 trackable debris fragments** and thousands of untrackable lethal shards that continue to threaten the ISS and operational satellites today (Kessler Syndrome).

Why Existing Screening Approaches Fail

1. **Computational Bottleneck:** Direct pairwise checking of N satellites requires $O(N^2)$ calculations, causing severe latency in operational pipelines.
2. **False-Alarm Fatigue:** Naive distance-only screening produces thousands of daily warnings without realistic collision probabilities (P_c), overwhelming operators.
3. **Manual Maneuver Delays:** Human-in-the-loop maneuver calculation takes hours when critical orbital decisions must be executed in minutes.

3 Slide 3: Solution Statement — How Space-Guard is Better

High-Level Solution (Non-Technical & Accessible)

Space-Guard is an autonomous, real-time orbital collision assessment and avoidance platform that watches the sky, detects close passes days in advance, and prescribes the most fuel-efficient escape burn.

How It Works in Three Clear Steps

- Continuous Orbital Ingestion:** Ingests live satellite tracking coordinates directly from global radar networks (CelesTrak / Space-Track).
- High-Speed Deterministic Screening:** Evaluates orbital trajectories through a fast two-stage mathematical filter, computing exact collision probability (P_c) in under 10 μ s.
- Optimal Avoidance Execution:** Calculates the exact thruster pulse (ΔV) required to steer the spacecraft away, using **14 times less fuel** by executing burns early.

Competitive Comparison Table

Feature / Capability	Traditional Legacy Tools	Space-Guard Platform
Screening Latency	Minutes to hours	< 5 seconds (Two-stage filter)
Collision Model	Geometric distance only	Analytic 2D Gaussian P_c (<10 μ s)
AI Role	Often none or black-box	Acceleration surrogate (No hallucinations)
Maneuver Optimization	Manual trial-and-error	Automated SVD CW Planner
Fuel Scaling	Late emergency burns	14× Early Secular Along-Track Drift
Real-World Validation	Synthetic data only	Verified on 2009 Iridium-Cosmos TLEs

Table 1: Comparison between Legacy Conjunction Tools and Space-Guard.

4 Slide 4: Implementation — Physics, Math & Computer Science

4.1 1. Physics and Mathematics

A. SGP4 Orbital Propagation

Converts Two-Line Element (TLE) mean Keplerian orbital elements $(a, e, i, \Omega, \omega, M)$ into instantaneous Cartesian state vectors $[\mathbf{r}, \mathbf{v}]$ in the Geocentric Celestial Reference System (GCRS/ECI non-rotating frame). SGP4 models non-spherical Earth gravitational harmonics (J_2, J_3, J_4) and atmospheric drag (B^*) .

B. Two-Stage Conjunction Screening

- **Stage 1 (Coarse Filter):** Filters all candidate pairs by checking overlap of apogee and perigee altitudes with a safety buffer:

$$r_{\text{perigee},i} - 50 \text{ km} \leq r_{\text{apogee},j} \quad \text{and} \quad r_{\text{perigee},j} - 50 \text{ km} \leq r_{\text{apogee},i}$$

This discards $> 90\%$ of non-threatening pairs in milliseconds.

- **Stage 2 (Fine TCA Search):** Applies `scipy.optimize.minimize_scalar` (Golden-Section / Brent minimization) to the scalar relative distance function:

$$f(t) = \|\mathbf{r}_i(t) - \mathbf{r}_j(t)\|$$

pinpointing the exact Time of Closest Approach (t_{TCA}).

C. Foster/Alfano 2D Gaussian Probability (P_c)

In the encounter B-plane normal to relative velocity $\mathbf{v}_{\text{rel}}$, the collision integral evaluates the 2D Gaussian probability density over the hard-body circle of combined radius $\text{HBR} = 10 \text{ m}$:

$$P_c \approx \frac{\text{HBR}^2}{2\sigma^2} \exp\left(-\frac{d_{\text{miss}}^2}{2\sigma^2}\right) \quad (1)$$

Because $\text{HBR} \ll \sigma$ ($\sigma = 500 \text{ m}$ assumed isotropic TLE uncertainty), probability density is uniform across the disk, yielding exact analytical computation in $< 10 \mu\text{s}$.

D. Clohessy-Wiltshire (CW) Maneuver Optimization

Linearized relative motion in near-circular orbit ($e < 0.01$) is governed by the state transition matrix:

$$\mathbf{r}(\Delta t) = \Phi_{rr}(\Delta t)\mathbf{r}_0 + \Phi_{rv}(\Delta t)(\mathbf{v}_0 + \Delta\mathbf{v})$$

The optimal burn direction is the right singular vector $\mathbf{v}_1$ from the Singular Value Decomposition of Φ_{rv} :

$$\Phi_{rv}(\Delta t) = U\Sigma V^T \implies \Delta\mathbf{v}_{\text{opt}} = \Delta V \cdot V_{[:,0]}$$

Along-track displacement accumulates via the secular term $\Delta y \approx -3n\Delta t\Delta v_y$, providing **14× more separation** at $\Delta t = 24\text{h}$ than at $\Delta t = 1\text{h}$.

4.2 2. Computer Science Architecture

- **Backend Microservices:** Asynchronous Python 3.12+ FastAPI server utilizing vectorized NumPy arrays and SciPy numerical optimizers.
- **Frontend Telemetry Workstation:** React 19 with Vite, Tailwind CSS, and Three.js WebGL canvas rendering 60 FPS 3D orbital dynamics.
- **Surrogate Acceleration Layer:** Pre-trained Random Forest classifier acting as an acceleration surrogate to pre-score pairs before deep propagation.

4.3 3. Satellite Data Acquisition Pipeline

- Direct automated fetch from **CelesTrak** / **Space-Track** active satellite catalogs.
- Local persistent cache with automated Time-To-Live (TTL) expiration to minimize external API rate limits.

5 Slide 5: Live Prototype Walkthrough Plan

The interactive prototype is demonstrated at <http://localhost:5173/demo> across four streamlined stages:

1. Stage 1: Live SGP4 Ephemeris Ingestion:

- Ingests live satellite catalog from CelesTrak.
- Displays real-time Cartesian GCRS coordinates $[X, Y, Z]$, velocities, and altitudes.

2. Stage 2: Physics-Based 3D Mapping:

- Visualizes real active satellites with accurate orbital inclination and nodal nodes on a 3D Earth globe.
- Demonstrates coarse altitude band filtering in 3D space.

3. Stage 3: 2009 Collision Reenactment:

- Replays the Iridium 33 vs Cosmos 2251 collision using real pre-collision TLEs (epoch 09041).
- Shows the head-on closing rate of 14.12 km/s, physical impact, and debris dispersion.

4. Stage 4: Avoidance Maneuver Verification:

- Interactive sliders demonstrating how an impulsive 0.10 m/s burn applied 24 hours prior achieves +4.83 km safe clearance, reducing P_c from 2.0×10^{-4} down to $< 10^{-6}$.

6 Slide 6: Target Audience & Societal Impact

- **1. National Space Agencies (e.g., ISRO, NASA, ESA):**
 - Directly augments ISRO's **IS4OM** (System for Safe and Sustainable Space Operations Management) at ISTRAC Bangalore.
 - Provides rapid preliminary triage for national assets (Cartosat, GSAT, RISAT, Chandrayaan/Aditya orbital paths).
- **2. Commercial Satellite Operators (e.g., Starlink, OneWeb, Pixxel, Skyroot):**
- Automated constellation screening and fuel-optimal avoidance to prolong spacecraft lifetime and preserve revenue.
- **3. Defence & Space Force Commands:**
- Autonomous surveillance of non-cooperative orbital objects and anti-satellite (ASAT) debris threats.
- **4. Space Insurance Underwriters:**
- Quantitative risk assessment and empirical collision probability auditing.

7 Slide 7: Development Roadmap

- **Phase 1: SIH Working Prototype (Current Completed Baseline):**
 - SGP4 propagation engine, two-stage screening filter, Foster/Alfano P_c model, CW maneuver optimizer, and 3D WebGL radar.
- **Phase 2: Full Catalog & Conjunction Data Messages (Q3–Q4 2026):**
 - Ingestion of Space-Track Conjunction Data Messages (CDMs) with full 3×3 error covariance matrices.
 - GPU-accelerated propagation scaling to the complete 27,000+ object catalog.
- **Phase 3: Multi-Satellite Constellation Autonomy (Q1–Q2 2027):**
 - Multi-agent avoidance de-confliction (ensuring maneuvers do not create secondary conjunctions).
 - Support for continuous low-thrust electric / ion propulsion trajectories.
- **Phase 4: Space Agency Protocol Integration (Q3–Q4 2027):**
 - Integration with CCSDS standards and direct telemetry handoff to mission operations centers.